

071306T4 EEN

ELECTRICAL ENGINEERING (POWER OPTION) LEVEL 6

ENG/OS/PO/CR/02/6

Install Electrical Power Lines

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**TVET CURRICULUM DEVELOPMENT, ASSESSMENT AND CERTIFICATION
COUNCIL (TVET CDACC)**

WRITTEN ASSESSMENT GUIDE

INSTRUCTIONS TO ASSESSOR

- i. *In this assessment, the candidate was required to answer **written** questions;*
- ii. *Marks for each question are indicated in the brackets*
- iii. *The paper consists of **TWO** sections: **A and B***

This guide consists of TEN printed pages.

Assessor should check the guide to ascertain that all the pages are printed as indicated and that no questions or answers are missing

SECTION A (40 Marks)

Note to assessor: These are suggested answers to act as guidelines

1. State four systems of power transmission. (4 marks)

- (i) **D.C. system**
- (ii) **Single-phase A.C. system**
- (iii) **Two-phase A.C. system**
- (iv) **Three-phase A.C. system**

Award @1 x 4 = 4 marks

2. State **four important components** of an overhead transmission line. (4 marks)

- (i) **Conductors** - which carry electric power from the sending end station to the receiving end station.
- (ii) **Supports** - which may be poles or towers and keep the conductors at a suitable level above the ground.
- (iii) **Insulators** - which are attached to supports and insulate the conductors from the ground.
- (iv) **Cross arms** - which provide support to the insulators.
- (v) **Miscellaneous** - items such as phase plates, danger plates, lightning arrestors, anti-climbing

Award @1 x 4 = 4 marks

3. Highlight **four** advantages of **pin-type insulator** used in power transmission lines (4 marks)

- i. **High Mechanical Strength**
- ii. **Simple Design**
- iii. **Ease of Installation**
- iv. **easier maintenance during the line's lifespan.**
- v. **Cost-Effectiveness.**
- vi. **The design of pin-type insulators, with their corrugated surfaces or petticoats, provides a longer leakage path for electrical current, enhancing insulation resistance and reducing the risk of flashovers, particularly in polluted environments.**
- vii. **Versatility: -Pin-type insulators come in various sizes and voltage ratings, making them adaptable to a wide range of transmission line applications. They can be used vertically or horizontally depending on the specific requirements of the line design.**

- viii. **Reliable Performance:** Their robust construction and inherent design features contribute to the overall reliability of pin-type insulators. They offer a long service life with minimal maintenance requirements, ensuring consistent and dependable performance of the transmission line.

Award @1 x 4 = 4 marks

4. Explain the **purpose of suspension insulators** as used in high voltage power transmission. (4 marks)

- (i) Suspension type insulators are cheaper than pin type insulators for voltages beyond 33 kV.
- (ii) Each unit or disc of suspension type insulator is designed for low voltage, usually 11 kV.
Depending upon the working voltage, the desired number of discs can be connected in series.
- (iii) If any disc is damaged, the whole string does not become useless because the damaged disc can be replaced by the sound one.
- (iv) The suspension arrangement provides greater flexibility to the line. The connection at the cross arm is such that insulator string is free to swing in any direction and can take up the position where mechanical stresses are minimum.
- (v) In case of increased demand on the transmission line, the additional insulation required for the raised voltage can be easily obtained in the suspension arrangement by adding the desired number of discs.
- (vi) The suspension type insulators are generally used with steel towers. As the conductors run below the earthed cross-arm of the tower, therefore, this arrangement provides partial protection from lightning.

Award @2 x 2 = 4 marks

5. **Outline three** disadvantages of wooden line support in transmission lines. (6 marks)

- (i) tendency to rot below the ground level
- (ii) comparatively smaller life (20-25 years)
- (iii) cannot be used for voltages higher than 20 kV
- (iv) less mechanical strength
- (v) require periodical inspection

Award @3 x 2 = 6 marks

6. Explain **two** methods of **reducing corona** effect in an overhead transmission line.

(4 marks)

- (i) **By increasing conductor size.** By increasing conductor size, the voltage at which corona occurs is raised and hence corona effects are considerably reduced.
- (ii) **By increasing conductor spacing.** By increasing the spacing between conductors, the voltage at which corona occurs is raised and hence corona effects can be eliminated.

Award @2 x 2= 4 marks

7. What is a **sag** in overhead lines?

(2 marks)

It is the difference in level between points of supports and the lowest point on the conductor.

Award 2 marks

8. Outline **two** factors which affect the sag of overhead transmission lines. (2 marks)

- i. **Weight:** Heavier conductors (due to material or diameter) naturally experience more sag due to their increased gravitational pull.
- ii. **Span length:** The distance between support towers significantly impacts sag. Longer spans allow for greater conductor dip due to less (tension).
- iii. **Tension:** the (tension) applied to the conductor directly affects sag. Higher (tension) pulls the conductor straighter, reducing sag. Lower (tension) results in more pronounced sagging.
- iv. **Wind:** Wind applies a lateral force on conductors, causing them to sway and potentially increasing sag in the wind direction.
- v. **Ice loading:** In areas prone to ice accumulation, the added weight of ice on conductors significantly increases their weight and leads to greater sag. This can be a critical design consideration.
- vi. **Temperature:** As mentioned earlier, temperature variations affect conductor material properties and influence sag.

Award @1 x 2= 2 marks

9. State three classifications of overhead transmission lines citing their voltage levels.

(6 marks)

- (i) **Short transmission lines.** When the length of an overhead transmission line is upto about 50 km and the line voltage is comparatively low (< 20 kV
- (ii) **Medium transmission lines.** When the length of an overhead transmission line is about 50- 150 km and the line voltage is moderately high (>20 kV < 100 kV),
- (iii) **Long transmission lines.** When the length of an overhead transmission line is more than 150 km and line voltage is very high (> 100 kV),

Award @2 x 3= 6 marks

10. Outline four **properties** of conductor material used for transmission and distribution of electric power. (4 marks)

- (i) **high electrical conductivity.**
- (ii) **high tensile strength in order to withstand mechanical stresses.**
- (iii) **low cost so that it can be used for long distances.**
- (iv) **low specific gravity so that weight per unit volume is small**

Award @1 x 4 = 4 marks

SECTION B (60 MARKS)

*Answer any **three** questions in this section*

11. (a) Describe the improvement of string efficiency by grading of units and guard ring. (6marks)

- (i) **By using longer cross-arms. The value of string efficiency depends upon the value of K i.e. The lesser the value of K, the greater is the string efficiency and more uniform is the voltage distribution. The value of K can be decreased by reducing the shunt capacitance.**
- (ii) **By grading the insulators. insulators of different dimensions are so chosen that each has a different capacitance.**
- (iii) **By using a guard ring. The potential across each unit in a string can be equalized by using a guard ring which is a metal ring electrically connected to the conductor and surrounding the bottom insulator.**

Award @2 x 3 = 6 marks

- (b) The three bus-bar conductors in an outdoor substation are supported by units of post type insulators. Each unit consists of a stack of 3 pin type insulators fixed one on the top of the other. The voltage across the lowest insulator is 13.1 kV and that across the next unit is 11 kV. Find the bus-bar voltage of the station. (14 marks)

Solution.

It is given that $V_3 = 13.1$ kV and $V_2 = 11$ kV.

Let K be the ratio of shunt capacitance to self-capacitance of each unit.

Applying Kirchhoff's current law to Junctions A and B, we can easily derive the following equations:

$$V_2 = V_1 (1 + K)$$

or $V_1 = \frac{V_2}{1 + K} \quad \dots(i)$

and $V_3 = V_2 + (V_1 + V_2) K \quad \dots(ii)$

Putting the value of $V_1 = V_2/(1 + K)$ in eq. (ii), we get,

$$V_3 = V_2 + \left[\frac{V_2}{1 + K} + V_2 \right] K$$

or $V_3 (1 + K) = V_2 (1 + K) + [V_2 + V_2 (1 + K)] K$

$$= V_2 [(1 + K) + K + (K + K^2)]$$

$$= V_2 (1 + 3K + K^2)$$

$\therefore 13 \cdot 1 (1 + K) = 11 [1 + 3K + K^2]$

or $11K^2 + 19 \cdot 9K - 2 \cdot 1 = 0$

Solving this equation, we get, $K = 0.1$.

$\therefore V_1 = \frac{V_2}{1 + K} = \frac{11}{1 + 0.1} = 10 \text{ kV}$

Voltage between line and earth $= V_1 + V_2 + V_3 = 10 + 11 + 13 \cdot 1 = 34 \cdot 1 \text{ kV}$

$\therefore$ Voltage between bus-bars (i.e., line voltage)

$$= 34 \cdot 1 \times \sqrt{3} = 59 \text{ kV}$$

(Award 1 mark for each correct step)

12. (a) Explain **three properties** of **line supports** in overhead transmission line (6marks)

- (i) **High mechanical strength to withstand the weight of conductors and wind loads etc.**
- (ii) **Light in weight without the loss of mechanical strength.**
- (iii) **Cheap in cost and economical to maintain.**
- (iv) **Longer life.**
- (v) **Easy accessibility of conductors for maintenance**

Award @2 x 3 = 6 marks

(b) Determine the efficiency and regulation of a 3-phase, 100 km, 50 Hz transmission line delivering 20 MW at a p.f. of 0.8 lagging and 66 kV to a balanced load. The conductors are of copper, each having resistance 0.1 ohm per km, 1.5 cm outside diameter, spaced equilaterally 2 meters between centres. Neglect leakage and use nominal-T, method.

(14marks)

Solution:

Total resistance of line $100 \times 0.1 = 10$ ohms.

$$\begin{aligned} \text{The inductance of the line} &= 2 \times 10^{-7} \times 100 \times 1000 \ln \frac{200}{0.75} \text{ H} \\ &= 11.17 \times 10^{-2} \text{ H} \end{aligned}$$

$$\therefore \text{Inductive reactance} = 314 \times 11.17 \times 10^{-2} = 35.1 \text{ ohm}$$

$$\begin{aligned} \text{The capacitance/phase} &= \frac{2\pi \times 8.854 \times 10^{-12}}{\ln \frac{200}{0.75}} \times 100 \times 1000 \\ &= 9.954 \times 10^{-7} = 0.9954 \text{ } \mu\text{F}. \end{aligned}$$

Nominal-T method: The nominal-T circuit for the problem is given below:

$$I_r = \frac{20 \times 1000}{\sqrt{3} \times 66 \times 0.8} = 218.68 \text{ amps}$$

$$V_r = \frac{66 \times 1000}{\sqrt{3}} = 38104 \text{ volts}$$

Taking I_r as the reference, the voltage across the condenser will be

$$\begin{aligned} V_c &= (38104 \times 0.8 + 218.68 \times 5) + j(38104 \times 0.6 + 218.68 \times 17.55) \\ &= 31576 + j26700 \end{aligned}$$

$$\begin{aligned} \text{The current } I_c &= j\omega CV_c \\ &= j314(31576 + j26700) \times 0.9954 \times 10^{-6} \\ &= j9.87 - 8.34 \\ \therefore I_s &= 218.68 + j9.87 - 8.34 = 210.34 + j9.87 \\ &= 210.57 \text{ amps} \end{aligned}$$

$$\begin{aligned} V_s &= V_c + I_s Z/2 \\ &= 31576 + j26700 + (210.34 + j9.87)(5 + j17.53) \\ &= 31576 + 1051 - 173 + j26700 + j3691 + j49.35 \\ &= 32454 + j30440 \end{aligned}$$

$$\therefore |V_s| = 44495 \text{ volts}$$

The no load receiving end voltage will be $(|V_s|(-j3199)) / (5 + j1755 - j3199) = 44746 \text{ volts}$

$$\therefore \% \text{ regulation} = \frac{44746 - 38104}{38104} \times 100 = \mathbf{17.4\%}. \text{ Ans.}$$

To determine η we evaluate transmission line losses as follows:

$$\begin{aligned} 3[218.682 \times 5 + 210.572 \times 5] &= 1382409 \text{ watts} = 1.3824 \text{ MW} \\ \therefore \% \eta &= \frac{20}{20 \times 13824} \times 100 = \mathbf{93.5\%}. \text{ Ans.} \end{aligned}$$

13. (a) Explain three importance of transposition in power transmission lines. (6 marks)

- i. **In order that voltage drops are equal in all conductors, we generally interchange the positions of the conductors at regular intervals along the line so that each conductor occupies the original position of every other conductor over an equal distance**

- ii. **Power transmission lines are transposed to prevent interference with neighboring telephone lines**
- iii. **It reduces the unbalance**
- iv. **The transposition arrangement of high voltage lines to reduce the system power loss**
- v. **Transposition arrangement of powerline helps to reduce the effect of inductive coupling**
- vi. **Transposition system for single circuit tower using same tension tower with reduced deviation angle**
- vii. **It is proved more economical**

Award @3 x 2 = 6 marks

(b) Highlight three advantages of Bundled - Conductor lines over Single- conductor lines of the same area of Cross section. (6 marks)

- i. **Lower inductance, which is beneficial for reducing transmission line impedance and improving power transfer efficiency.**
- ii. **Increased current carrying capacity due to the larger overall area.**
- iii. **Reduced corona discharges due to the increased effective radius**

Award @2 x 3 = 6 marks

(c) Describe **four** types of tests done on transmission lines. (8 marks)

- i. **Continuity Test**
The continuity test, also known as the conductor continuity test, is conducted to ensure that all conductors (wires or cables) within a transmission line are correctly connected and that there are no open or broken circuits.
- ii. **Short Circuit Test**
The short circuit test, also known as a fault current test, is designed to assess how well the transmission line can handle and clear a short circuit fault safely.
- iii. **Earth Continuity Test**
The earth continuity test, also referred to as the ground continuity test, is performed to ensure that the electrical equipment and conductive parts of a transmission line are effectively grounded.
- iv. **Insulation Test**
The insulation test assesses the condition of the insulation material in the transmission line to ensure it can withstand the voltage stresses it will encounter during normal operation.

(Award 2 marks to each)

14. (a) Outline three assumptions made when deriving sag equation for equal supports.

(6 marks)

- i. The conductor is assumed to have a constant weight per unit length.
- ii. conductor can deform freely under tension without any change in its length.
- iii. The conductor's cross-section is assumed to be circular for simplicity of analysis.
- iv. The supports at both ends of the span are assumed to be at the same height.
- v. The supports offer no frictional resistance to the movement of the conductor.
- vi. The conductor is assumed to be subjected only to its own weight and the tension induced by its sag.
- vii. The temperature along the conductor is assumed to be constant.

Award @2 x 3 = 6 marks

(b) A transmission line conductor having a diameter of 19.5 mm weights 0.85 kg/m. The span is 275 metres. The wind pressure is 39 kg/m² of projected area with ice coating of 13 mm. The ultimate strength of the conductor is 8000 kg. Calculate the maximum sag if the factor of safety is 2 and ice weighs 910 kg/m³.

(14 marks)

Solution:

The overall diameter of the conductor with ice coating = 1.95 + 2.6 = 4.55 cm

The projected area per metre length of conductor = $4.55 \times 1 \times 10^{-2}$ sq. metre

The wind load per metre length = $4.55 \times 10^{-2} \times 39 = 1.77$ kg/metre

The area of section of ice

$$\begin{aligned}
 &= \pi[(r + t)^2 - r^2] \\
 &= \pi[22.75^2 - 9.75^2] \\
 &= \pi(517.56 - 95.06) \\
 &= 1327.32 \text{ sq. mm} \\
 &= 1.327 \times 10^{-3} \text{ sq. metre}
 \end{aligned}$$

∴ Ice weight per metre length = $1.327 \times 10^{-3} \times 910 = 1.207$ kg/metre

The total weight W

$$\begin{aligned}
 &= \sqrt{(0.85 + 1.207)^2 + 1.77^2} \\
 &= \sqrt{4.23125 + 3.1329} \\
 &= 2.71369 \text{ kg/metre}
 \end{aligned}$$

The working tension of the conductor = $\frac{8000}{2} = 4000$ kg

$$\therefore c = \frac{T}{W} = \frac{4000}{2.71369} = 1474 \text{ metres}$$

$$\therefore \text{The maximum sag} = \frac{l^2}{8c} = \frac{275^2}{8 \times 1474} = 6.4 \text{ metres.}$$

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